Exam in Simulation Methods in Medical Engineering, CM2014

Monday May 30th 2022, 14.00 – 18.00

You may not use aids such as books, notes, text or images.

Basic-Level / E-level

Mathematical and numerical background

1. Given the ODE-problem

$$\frac{dx}{dt} = -xt + 1, \quad x(0) = 1 \qquad (1)$$

a) Formulate Euler's explicit method for the problem and calculate the approximation of $x(1)$ with the stepsize $h = 0.5$ 2 p

b) Formulate Euler's implicit method for the problem and calculate the first step forward with the stepsize $h = 0.5$ 2 p

2. Given the ODE-problem

$$\frac{d^2y}{dx^2} + xy = 1, y(0) = 1, y(1) = 0 \qquad (2)$$

a) Is this an initial or a boundary value problem? 1 p

b) Discretize (2) with the finite difference method (FDM) the problem on a grid $x_i = ih, i = 0,1,2, \dots, N$ where $hN = 1$. Let $N = 4$. Formulate the linear system of equations $Ay = b$ obtained with the FDM, i.e. find and write out the matrix A and the right hand side b. 2 p

Finite Element Method (FEM)

3.

a) Identify, and briefly explain the potential sources of error in finite element analysis 2 p

b) Give at least two examples of engineering of biomechanics problems that can be solved by finite element analysis and briefly describe your approach of creating the model. What is needed? 3 p

SDM, DES, ABM combined

4. Modelling and simulation is commonly used in epidemiology to understand population data on the spread of infectious disease. The types of models used depend on the purpose of the analysis, this may for example include system dynamic models, discrete event simulation, or agent-based models.

   You work as a modeller at a public health agency. You have been tasked with modelling the outbreak of a novel viral disease.

   a) As the in-house modelling expert - explain what system thinking is and how system dynamics fit into this paradigm. Exemplify this using a simple model of spread of disease (you can choose if you would like to explain this in words, a diagram or equation system). 3 p

   b) Model development relies on several activities in order for users and decision-makers to build confidence in the approach. Describe model verification and validation. Use examples on how you could carry out this approach for the above model. 3 p

Advanced-Level

Mathematical and numerical background

5. Consider the ODE-problem

$$\frac{d^2y}{dt^2} + y = 0, \;\; y(0) = 1, \frac{dy}{dx}(0) = 0 \qquad (5a)$$

a) Verify that the solution is $y = \cos(t)$ and that the total energy $H(t) = y^2 + \left(\frac{dy}{dt}\right)^2$ is constant for all $t \geq 0$ 1 p

b) Show that (5a) can be given on matrix form

$$\frac{du}{dt} = Au, u(0) = u_0 \qquad (5b)$$

where A is a 2x2 matrix and u is a 2x1 vector. What is the numerical value of A and the initial vector u_0? 2 p

c) When a numerical method is applied to (5b) a sequence of vector values $u_0, u_1, u_2, \ldots . u_k, u_{[k+1]}, \ldots \ldots \ldots$ is obtained. From these values approximations $H_k = u_k^T u_k$ of the energy can be calculated. When Euler's explicit method is used will the H_k-values be constant, i.e. will the energy be preserved? 1 p

d) When the Implicit Midpoint Method (IMM) is applied to (5b)

$$u_{[k+1]} = u_k + \frac{h}{2}(Au_{[k+1]} + Au_k)$$

verify that the energy values H_k obtained with IMM will be constant. Hint: verify that $u_{[k+1]}^T u_{[k+1]} = u_k^T u_k$ for $k = 0,1,2, \ldots$ 2 p

Finite Element Method (FEM)

6.

a) Describe the difference between dynamic and static FEM. Indicate the parts in the equation that are negligible in static FEM and motivate why they can be assumed to be negligible. Approximately at which deformation rate is it required to switch from static to dynamic FEM? 2 p

b) A cell-phone manufacturer uses dynamic FEA to ensure that the cell-phones manage to be dropped from two meters without breaking. Would you use explicit or implicit analysis to solve this problem? Motivate your answer! 2 p

c) Later, the same manufacturer wants to know if the cell-phone can stand a heavy person stepping on it after the fall. Assume that the loading lasts a couple of seconds. Would you use explicit or implicit analysis to solve this problem? Motivate your answer! 2 p

SDM, DES, ABM combined

7. *This is a continuation of the basic level question 4:*
   Same background as in question 4, the question below:

   The epidemiological experts are contemplating on how additional data and detail could inform the epidemiological model. What additional insights could you gain going to an agent-based model as compared to system dynamic modelling? What is the impact on the data requirements? 4 p

Good luck!